

Enhancing tourism digital twins with real time Bilateral data across Sri Lanka

Group ID = 25-26J-380

Git repo link -

<https://github.com/ChamothAshen/Enhancing-tourism-digital-twins-with-real-time-Bilateral-data-across-sri-lanka>

SIGIRIYA TOURISM DIGITAL TWIN PLATFORM

An Integrated AI-Powered Tourism Management System for Sigiriya Rock Fortress, Sri Lanka

1. PROJECT OVERVIEW

The **Sigiriya Tourism Digital Twin Platform** is a comprehensive AI-driven system designed to enhance tourism management at the **Sigiriya Rock Fortress**, a UNESCO World Heritage Site.

This platform integrates **artificial intelligence, computer vision, predictive analytics, environmental risk assessment, and mobile technologies** to provide real-time monitoring, forecasting, and intelligent guidance for tourists and authorities.

2. KEY OBJECTIVES

- Real-time crowd monitoring and people counting
- Visitor forecasting and demand prediction
- Environmental risk assessment (fog, heat stress, slip risk)
- AI-powered mobile tour guidance
- Data-driven insights for sustainable tourism management

3. PROJECT STRUCTURE

50percent/	
├── Croud/TourismDigitalCC/	Crowd Detection Module
├── FC/TourismDigitalFC/	Forecasting & Chat API
├── 3d/Risk Prediction/	Environmental Risk Prediction
├── Location guide/path_predict/	Movement & Path Prediction
└── sigiriya_tour_guide/	Flutter Mobile Application

4. SYSTEM MODULES

4.1 Crowd Detection System (YOLOv8)

Technology Used:
YOLOv8 (You Only Look Once – Version 8)

- Main Features:**
- Real-time person detection
 - Crowd counting and density monitoring
 - Batch image and video frame processing
 - Adjustable confidence thresholds
 - Visual bounding box annotations
 - Statistical summaries and reports

- Use Cases:**
- Monitor congestion at entry points
 - Detect overcrowding risks
 - Support safety and compliance monitoring
 - Analyze historical crowd patterns

- Model Details:**
- Model: YOLOv8 Nano
 - Size: ~6 MB
 - Performance: Real-time capable
 - Person class ID: 0
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4.2 Visitor Forecasting System (Prophet + FastAPI)

Technology Used:

Facebook Prophet, FastAPI, Python

Capabilities:

- 90-day visitor forecasting
- Best date recommendations
- Day-of-week and seasonal analysis
- Weather and holiday impact consideration
- REST API and AI chat interface

API Endpoints:

- /forecast
- /recommendations
- /chat
- /best-dates
- /day-patterns
- /health

Key Features:

- Weekly and yearly seasonality detection
 - Festival season analysis
 - Confidence interval prediction
 - Interactive web interface
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4.3 Environmental Risk Prediction

Technology Used:

Scikit-learn Machine Learning Models

Risk Types:

Fog Risk

- Visibility assessment
- Climbing safety alerts

Heat Stress Risk

- Temperature and humidity analysis
- Visitor comfort evaluation

Slip Risk

- Rainfall and surface condition analysis
- Safety warning generation

Input Parameters:

- Temperature
- Humidity

- Wind speed
 - Rainfall
 - Cloud coverage
 - Visitor count
 - Time of day
-

4.4 Movement & Path Prediction

Technology Used:

Multi-class classification models

Features:

- Tourist movement prediction
- Popular route identification
- Congestion hotspot detection
- Optimal path recommendations

Models Applied:

- Random Forest
 - Logistic Regression
 - Gradient Boosting
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4.5 Sigiriya Tour Guide Mobile Application

Technology Used:

Flutter (Cross-platform)

Key Features:

- Google Maps integration
- Real-time GPS tracking
- 8 major Points of Interest (POIs)
- Proximity-based notifications
- Distance calculations
- Animated navigation

Supported Platforms:

Android, iOS, Web, Windows, macOS, Linux

5. INSTALLATION AND SETUP

System Requirements

- Python 3.8 or higher
 - Flutter SDK 3.0 or higher
 - Minimum 4 GB RAM
 - 5 GB free disk space
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6. DATA SOURCES

Visitor Data

File: sigiriya_synthetic_visitors_2023_2025.csv

Key Attributes:

- Date
 - Average temperature
 - Rainfall
 - Public holidays
 - Festival season
 - Visitor count
-

Microclimate Data

File: sigiriya_synthetic_microclimate.csv

Key Attributes:

- Temperature
 - Humidity
 - Wind speed
 - Rainfall
 - Fog risk
 - Slip risk
 - Heat stress
-

7. MODEL PERFORMANCE SUMMARY

Crowd Detection:

- Accuracy: Above 90%
- Real-time inference capability

Visitor Forecasting:

- Mean Absolute Error: ~250 visitors
- Forecast horizon: 90 days

Risk Prediction Models:

- Fog risk accuracy: 85–90%
 - Heat stress accuracy: 88–92%
 - Slip risk accuracy: 83–87%
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8. DEPLOYMENT

Backend Deployment:

- Docker containerization
- Cloud platforms (AWS, Azure, GCP)

Mobile App Deployment:

- Android APK / App Bundle
 - iOS build
 - Web deployment
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9. FUTURE ROADMAP

Phase 1 – Completed

- Crowd detection
- Forecasting API
- Risk prediction
- Mobile tour guide

Phase 2 – Planned

- Real-time dashboard
- Push notifications
- Admin management panel

Phase 3 – Future Enhancements

- AR/VR tourism experiences
 - Voice-guided tours
 - Multi-language support
 - Multi-site expansion
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10. CONCLUSION

The **Sigiriya Tourism Digital Twin Platform** demonstrates how AI and digital twin technologies can be applied to **heritage site management**, improving safety, visitor experience, and sustainability. This system provides a strong foundation for future smart tourism innovations in Sri Lanka and beyond.

11. LICENSE

This project is licensed under the **MIT License**.

12. ACKNOWLEDGMENTS

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- Facebook Research – Prophet
- FastAPI Team
- Google Maps & Flutter
- UNESCO – Sigiriya World Heritage Site
- Sri Lanka Tourism Authority

SYSTEM DIAGRAM

